

IE 544 – Decision Analysis, Assignment 3

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1 Generative AI Acknowledgment

We used Generative AI models such as Claude and Gemini, to use some functions in R like `tapply` and `ave()` functions. Also, we heavily used those models to outline our report.

2 Summary of our Study

We study which factors decide the winner of Amazon’s Buy Box using a fully discrete Bayesian Network learned with the `bnlearn` package. The data describe 130,278 seller offers across 9 products and 184 sellers, sampled between 11 August and 2 September 2015. The central finding is that the Buy Box is won mainly by Amazon itself, and, among third-party sellers, by the seller offering the lowest price while clearing minimum rating and fulfilment requirements. We build a network that reproduces these patterns and test its predictions on test data.

3 Question 1: Descriptive Analysis

Number of Products, number of sellers (1a) Each row is one seller’s offer for one product at one timestamp, so several sellers compete in each snapshot. The training data contain 9 products, 184 sellers, 130,278 offer-rows, and 1,117 snapshots. Across all rows the Buy Box is won only 6.9% of the time, which makes sense because each snapshot has roughly one winner among many competing offers.

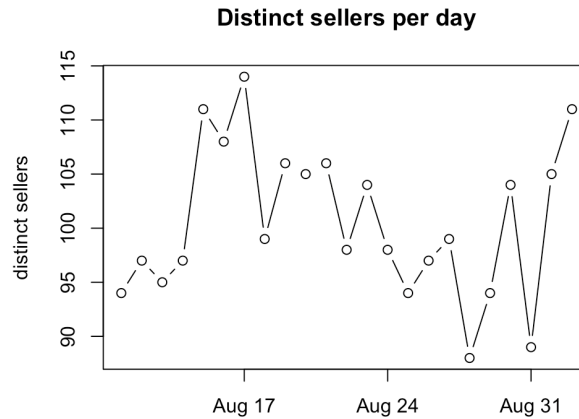


Figure 1: Distinct Sellers Per Day

Prices and ratings by product (1b, 1c) Price levels differ widely across the 9 products, so absolute price is not comparable between products which motivates the relative price feature in 1f.

Table 1 should be pasted from your `price_by_pid` and `ratings_by_pid` outputs.

Table 1: Price and shipping summary by product (training data, price > 0).

Product (ASIN)	Min	Mean	Max	Mean BB price	Mean shipping
B002ZV0OJO	163.80	282.27	1122.41	221.64	5.20
B0083H1INK	209.99	331.94	1271.56	255.15	0.57
B00AMFLZLG	259.99	317.93	852.47	259.99	1.37
B00DNSO1OW	88.39	159.85	951.43	88.39	2.26
B00DNSO41M	197.00	370.33	1171.72	291.35	5.05
B00MVVI1FC	74.99	181.53	1018.94	86.96	6.13
B00VSIT5UE	975.71	997.45	999.00	999.54	0.41
B00VSITBJO	793.01	1253.11	1259.00	1256.84	1.19
B00YR6BMS2	700.00	997.06	1633.60	815.81	4.68

Table 2: Seller ratings and feedback by product (training data).

Product (ASIN)	Mean sid_rating	Mean pos_fb	Mean rating_cnt	pid_rating
B002ZV0OJO	4.46	8.80	5.98	4.0
B0083H1INK	4.52	8.95	5.70	4.0
B00AMFLZLG	2.68	5.29	0.03	4.5
B00DNSO1OW	4.49	8.87	8.16	4.5
B00DNSO41M	4.20	8.04	2.50	4.5
B00MVVI1FC	4.77	9.57	0.25	4.5
B00VSIT5UE	4.87	9.60	3.70	4.5
B00VSITBJO	4.46	8.73	5.44	4.5
B00YR6BMS2	4.63	9.16	6.53	4.5

Amazon vs. other sellers (1d) When Amazon is the seller it wins the Buy Box **92.6%** of the time; every other seller combined wins only **2.9%** of the time (Table 3).

Table 3: Buy-Box win rate, Amazon vs. all other sellers (training data).

Seller type	Win rate	# offers
Amazon	0.926	5,854
Other	0.029	124,424

Prime, FBA, page and rank (1e) Fulfillment by Amazon matters only through Prime. The offers that are FBA but not Prime win essentially **0%** of the time, while FBA-and-Prime offers win **30.4%** (Table 4). Position also matters strongly. An offer at the very top of page 1 wins about **61%**, whereas the top of page 2 wins only **0.3%**, which is an evidence that page and rank should be combined into a single global rank.

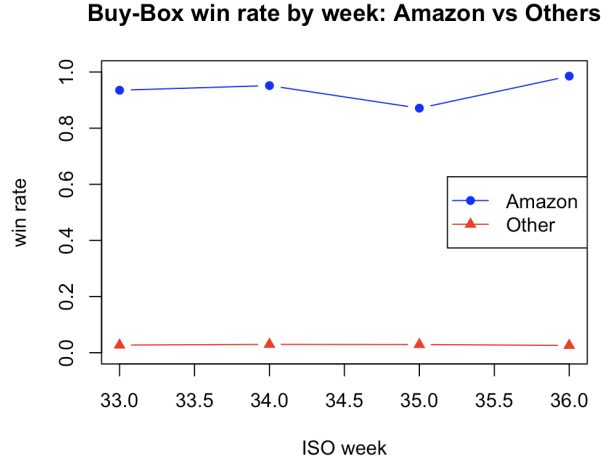


Figure 2: Win Rate by week, Amazon vs. Others

Table 4: Win rate by fulfilment status.

FBA	Prime	Win rate	# offers
no	no	0.032	87,708
yes	no	0.000	22,167
yes	yes	0.304	20,403

New price feature (1f) Because absolute price is not comparable across products, we define, within each product-snapshot, the *price premium* = (this offer’s price) / (lowest price in the snapshot). A value of 1.0 means the offer is the cheapest. The win rate falls sharply as the premium grows (Table 5), which is the empirical justification for the price bins used later.

Table 5: Win rate by price-premium bin (training data).

Price premium	Win rate	# offers
cheapest	0.512	10,358
< 5% over	0.051	34,941
5–15% over	0.034	21,536
> 15% over	0.019	63,443

Measurement scales (1g) Table 6 lists the scale of each variable. There are two notes we want to bring up here. The data dictionary says `sid_pos_fb` is on a 0–100 scale, but in the actual data it ranges 0–10; and `pid_rating` only ever takes the values 4.0 or 4.5, so it carries almost no information.

Table 6: Measurement scale of each variable.

Variable	Scale	Note
pid, sid, bbox_sid	nominal	identifiers / labels
epoc	interval	timestamp
price, shipping, bbox_price	ratio	true zero
sid_rating, pid_rating	interval/ratio	bounded 0–5
sid_pos_fb	ratio	bounded; 0–10 in data
sid_rating_cnt	ratio	scaled 0–100
pid_rating_cnt	ratio	scaled 0–1
page, rank	ordinal	ordered positions
is_fba, is_prime, bbox	nominal	binary

Outliers (1h) A boxplot of price by product flags some very high prices. We inspected the extreme rows (more than $5\times$ or less than $1/5$ of the product median) and found that none of them ever won the Buy Box. We decided to keep them because a genuine high price that never wins is informative about how losing looks. We discard only clear scraping artefacts (`price = 0`), as the data dictionary advises.

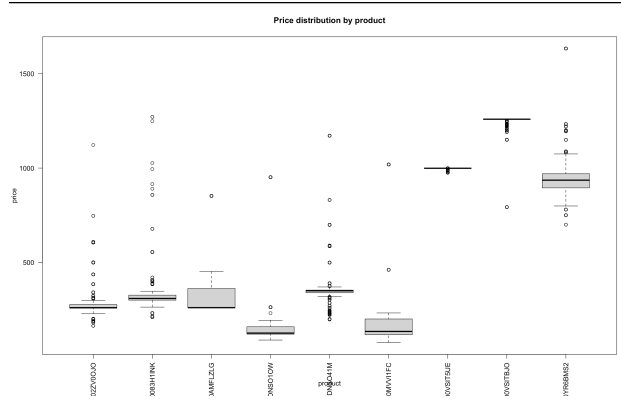


Figure 3: Boxplot of price by product

4 Question 2: Interpretation and Graphical Model

The descriptive results tell a consistent story about how a seller wins the Buy Box:

- **Being Amazon** is by far the strongest signal (92.6% vs. 2.9%).
- **Fulfilment** matters only through **Prime**; FBA alone is not enough.
- Among third-party sellers, **price** is the operative lever: even after removing Amazon, the cheapest offer wins 26% of the time versus about 1% otherwise – so the price effect is real and not merely an Amazon artefact.
- **Seller credibility** (rating, positive feedback, rating count) acts as a **filter**: below about 4 stars or below 7/10 positive feedback the win rate is essentially 0%.

- **Position** (combined page+rank) is strongly associated with winning.

We also observe structural relationships *between* the predictors: Amazon offers are always FBA and always Prime, and a Prime offer is always FBA. These become directed edges and constraints in the network.

Proposed DAG The hypothesised graphical model places seller identity, price, credibility and position as causes that flow into the outcome **bbox**, with fulfilment mediating between seller identity and the outcome.

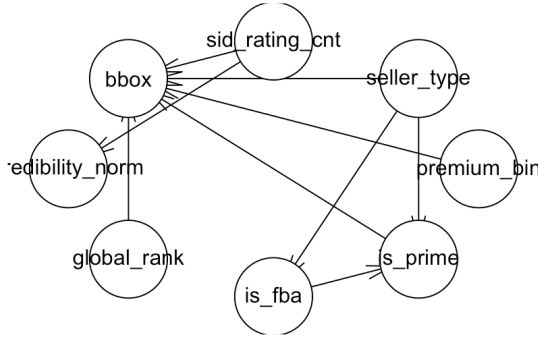


Figure 4: Hypothesized DAG

Whitelist / blacklist We force the structural certainty $\text{is_fba} \rightarrow \text{is_prime}$ (whitelist). We forbid (blacklist): any edge *out of* **bbox** (the outcome causes nothing upstream), any edge *into* **seller_type** (seller identity is fixed beforehand), and $\text{is_prime} \rightarrow \text{is_fba}$.

Position rank and the Buy Box may both be consequences of the same hidden Amazon ranking algorithm. We model $\text{rank} \rightarrow \text{bbox}$ because rank is observable before the assignment, but this arrow is causally debatable and we flag it as such.

5 Question 3: Structure Learning

The outcome is categorical and several predictors are already categorical; moreover, the price effect is threshold-like. Being cheapest matters, but not the dollar amount. A fully discrete network captures such non-linear effects, so we discretise the remaining continuous variables. Bin edges were chosen from the Q1 win-rate tables (e.g. the price-premium staircase) and from quantiles for the count-based variables. The modelled variables are: **seller_type**, **is_fba**, **is_prime**, **premium_bin**, **sid_rating_cnt.bin**, **credibility_norm.bin**, **global_rank_bin**, and the outcome **bbox**. We drop **pid_rating** (no information), **shipping** (missing values), and the Buy-Box columns (forbidden for prediction).

Note on global rank. Because **global_rank_bin** tops out at rank 11, every page-2 offer is dropped; the model is therefore built on **page-1 offers (about 63% of rows)**. This loses almost no winners, since page-2 offers essentially never win, but it should be stated explicitly.

Three algorithms. We learned the structure with a score-based method (hill-climbing, BIC), a constraint-based method (IAMB), and a hybrid method (MMHC), all under the same whitelist/blacklist. Hill-climbing searches for the graph with the best score (fit minus a complexity penalty); IAMB keeps only edges that survive conditional-independence tests; MMHC finds the skeleton by tests and orients it by score. The results are compared in Table 7; the three structures are fairly similar (structural Hamming distance: hc-iamb 16, hc-mmhc 20, iamb-mmhc 8).



Figure 5: Hill climbing

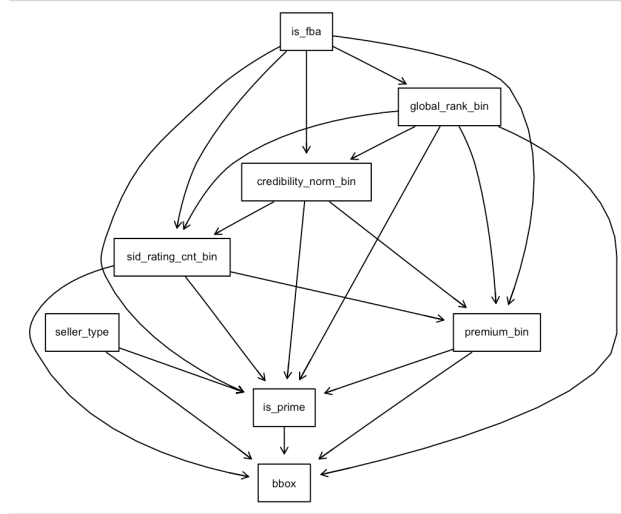


Figure 6: IAMB

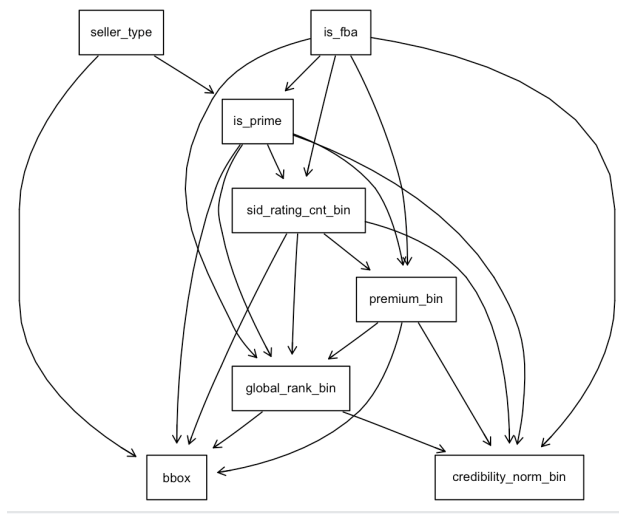


Figure 7: MMHC

Table 7: Comparison of the three learned structures.

	Hill-climbing	IAMB	MMHC
# arcs	25	21	21
BIC (higher better)	−607,437	−626,486	−622,169
10-fold CV error (lower better)	0.0134	0.0134	0.0164

Robustness (bootstrap). We resampled the data 100 times and re-learned the structure each time. The arcs that pointed *into* the outcome appeared with strength 1.00 and a consistent direction in every resample (Table 8), so the set of Buy-Box drivers is highly stable: seller identity, fulfilment, price, credibility count, and position.

Table 8: Bootstrap arcs into `bbox` (strength and direction over 100 resamples).

From	To	Strength	Direction
<code>seller_type</code>	<code>bbox</code>	1.00	1.00
<code>is_fba</code>	<code>bbox</code>	1.00	1.00
<code>premium_bin</code>	<code>bbox</code>	1.00	1.00
<code>sid_rating_cnt_bin</code>	<code>bbox</code>	1.00	1.00
<code>global_rank_bin</code>	<code>bbox</code>	1.00	1.00

Model choice. We chose the **hill-climbing** model. It has the best BIC by a wide margin (−607,437 vs. −622,169 for MMHC and −626,486 for IAMB), the (tied-)lowest cross-validation error (0.0134, equal to IAMB and better than MMHC’s 0.0164), and its arcs into the outcome are all bootstrap-stable. In the chosen network the direct parents of `bbox` are `seller_type`, `is_fba`, `premium_bin`, `sid_rating_cnt_bin` and `global_rank_bin`; `is_prime` and the credibility score influence the outcome indirectly, earlier in the chain.

6 Question 4: Parameters, Inference and Prediction

Parameter learning (4a). We fitted the conditional probability tables with `bn.fit` using the Bayesian method (a light Dirichlet prior, `iss=1`), which smooths cells that have no observations so they do not receive probability exactly zero. The fitted table for `bbox` is large (it has five parents), but reading it confirms the story: for example, an Amazon, FBA, cheapest, top-quartile-count offer wins with probability close to 1, while a non-cheapest non-Amazon offer wins with probability near 0.

Probability that Amazon wins (4b). Querying the fitted network for $P(\text{bbox} = \text{success} \mid \text{seller_type} = \text{Amazon})$ gives **0.9268**, essentially identical to the empirical training value of **0.9264** (gap 0.0004). The near-perfect match shows the network has correctly absorbed the dominant Amazon effect.

Prediction method (4c). We predict `bbox` with `method = "bayes-lw"` (likelihood weighting). Unlike prediction from parents only, this uses *all* observed variables, so it extracts more information from each test row – which is why the assignment hints that “prediction from parents may not be the best option.” The model uses no Buy-Box-derived variable, as required.

Accuracy on test data (4d). We applied the identical feature pipeline to `amz_test_full.rds` (41,854 offers). The confusion matrix is in Table 9 and the metrics in Table 10.

Table 9: Confusion matrix on the test data (chosen model: hill-climbing).

Predicted	Actual	
	success	failure
success	1,480	1,339
failure	1,318	37,717

Table 10: Test-set performance (chosen model).

Metric	Value
Overall accuracy	0.9365
Balanced accuracy	0.7473
Sensitivity (recall on success)	0.5289
Specificity (recall on failure)	0.9657
Precision (success)	0.5250

Model comparison on the test set. Re-fitting all three structures and scoring them on the test data (Table 11) shows they perform almost identically: hill-climbing has the highest raw accuracy (0.9365), while IAMB and MMHC edge it on balanced accuracy (0.7509 vs. 0.7473) by a fraction of a point. Combined with its best BIC and CV results, this confirms hill-climbing as a sound choice.

Confusion matrix — chosen model (HC)

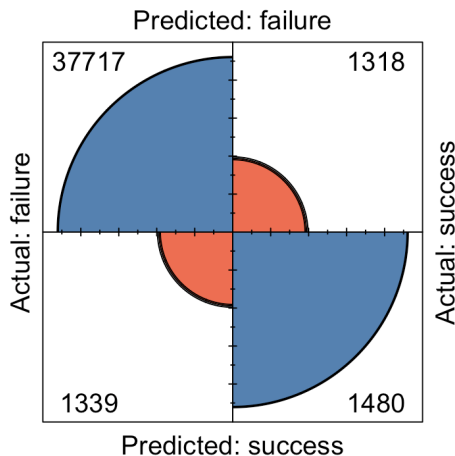


Figure 8: Confusion Matrix

Table 11: Test-set performance by algorithm.

	Hill-climbing	IAMB	MMHC
Accuracy	0.9365	0.9339	0.9339
Balanced accuracy	0.7473	0.7509	0.7509

Interpretation: distribution shift. The most informative result is in the errors. Amazon won 92.6% of its offers in training but only about **60%** in the test period. The model, calibrated to the training base rate, therefore over-predicts “success” for Amazon offers: the false positives (1,339) almost equal the true positives (1,480), and sensitivity (0.53) is far below specificity (0.97). In words, the model is very reliable at identifying losers but catches only about half of the actual winners, because many Amazon offers it confidently expects to win actually lost in the new period. The model captures the correct *mechanism* but is miscalibrated to the new period’s *base rate* – a reminder that Bayesian Networks assume the training and test distributions are similar.

7 Question 5: Conclusion

The picture is consistent across every part of the analysis. The strongest factor is simply **being Amazon**. For third-party sellers, the Buy Box behaves like a gated price competition: a seller must first clear **credibility** and **fulfilment** thresholds, and among those who qualify, the **lowest price** (and, relatedly, the top **position**) wins most often. Price alone is not sufficient and not the only factor, exactly as Amazon states. The bootstrap confirms that seller identity, fulfilment, price, credibility, and position are the stable drivers of the outcome.

They are a good fit for three reasons: the outcome and most drivers are categorical; the network makes the dependence structure explicit and interpretable (we can read off which factors matter and query conditional probabilities directly, as in 4b); and confounding checks, such as “is the price effect just an Amazon effect?”, are answered naturally by conditioning. The main limitations are that discretisation discards some detail, that the model assumes the test distribution matches training

(violated here, as Amazon’s win rate dropped from 93% to 60%, pulling test sensitivity down to 0.53), and that an observational network cannot by itself settle causal direction for ambiguous arrows such as $\text{position} \rightarrow \text{bbox}$. Overall, the Bayesian Network gives a clear, defensible description of the Buy-Box mechanism while honestly exposing where its assumptions are strained.